



SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

PROF. ARINDAM BANERJEE
Department of Mathematics
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Module 06: Regression

Lecture 01: Linear Regression 2



Linear Modelling

Setup:

- 1 Independent variables: $X_1, \dots, X_k$.
- 2 Dependent variable: Y .
- 3 $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \varepsilon$ where $E(\varepsilon) = 0$ and $\text{Var}(\varepsilon) = \sigma^2$

Goal: To model the above using i.i.d. sample of n data points from the above population $(X_{11}, \dots, X_{1k}, Y_1), \dots, (X_{n1}, \dots, X_{nk}, Y_n)$. So, for all i ,

$$Y_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_k X_{ik} + \varepsilon_i$$

where $E(\varepsilon_i) = 0$ and $\text{Var}(\varepsilon_i) = \sigma^2$.

The linear model

- Consider a sample linear regression model

$$Y = a + bX + \varepsilon$$

where X is the independent or explanatory variable and Y is the dependent or study variable.

- The terms a and b are the parameters of the model.
- The parameter a term is termed as intercept.
- The parameter b is termed as the slope parameter.
- The unobservable error component ε accounts for the failure of data to lie on a straight line and represents the difference between the true and observed realization of Y .

The linear model (continued)

- The independent variables are non-stochastic where as Y is viewed as random variable with

$$E(Y) = a + bX$$

$$E(\varepsilon) = 0$$

$$\text{and } \text{Var}(Y) = \sigma^2.$$

- If X is a random variable, then instead of the sample mean and sample variance of Y , we consider the conditional mean of Y given $X = x$ as

$$E(Y|X = x) = a + bX$$

and the conditional variance of Y given $X = x$ as

$$\text{Var}(Y|X = x) = \sigma^2$$

- In order to know the values of these parameters, n pairs of observations (X_i, Y_i) , $i = 1, 2, \dots, n$ on (X, Y) are observed/collected and are used to determine these unknown parameters.

Least squares estimation

- Suppose a sample of n sets of paired observations $(X_i, Y_i), i = 1, \dots, n$ is available.
- These observations are assumed to satisfy the simple linear regression model,

$$Y_i = a + bX_i + \varepsilon_i$$

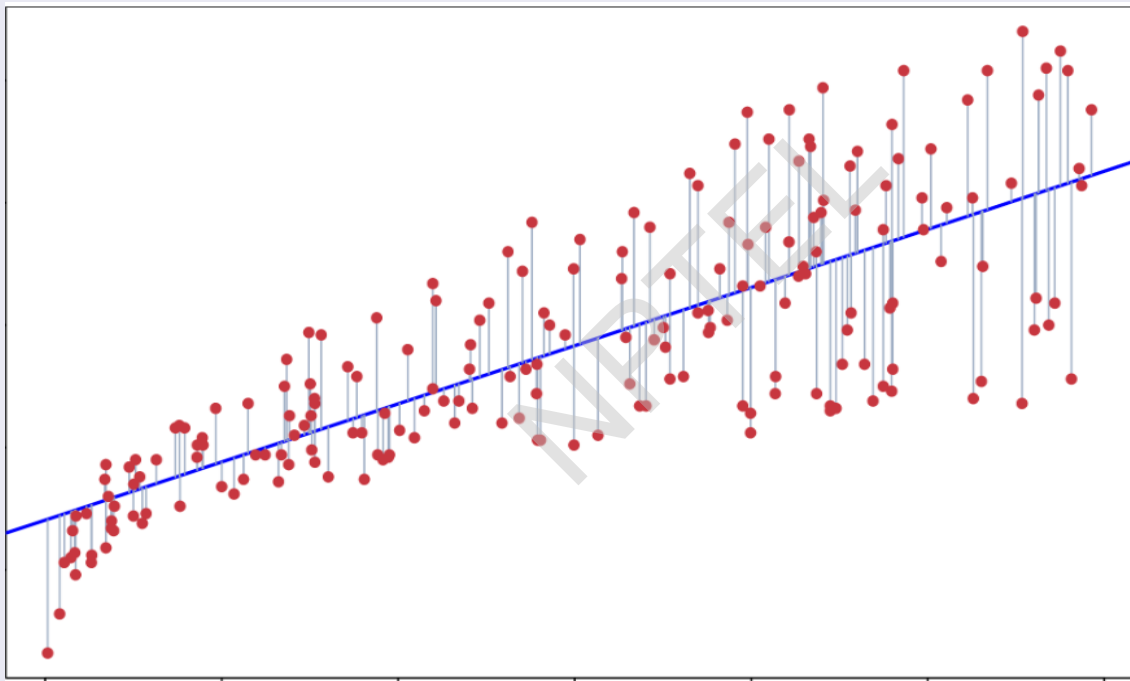
for $i = 1, 2, \dots, n$.

- The principle of least squares estimates the parameters a and b by minimizing the sum of squares of the difference between the observed and the predicted values of Y .

Regression

- Regression is a statistical method used to study the relationship between a dependent variable (also called the response variable) and one or more independent variables (also called predictor or explanatory variables).
- In regression analysis, we try to model how the dependent variable changes when one or more independent variables change, while keeping the others fixed.

A simple picture of regression



Linear Regression

- *Sample*: Collection of n data points $(X_1, Y_1), \dots, (X_n, Y_n)$.
- Want to find a nice function, $Y = f(X)$.
- One decent guess of f : the least square line !!
- Least Square Line: Line $Y = \hat{a} + \hat{b}X$ such that $a = \hat{a}$ and $b = \hat{b}$ minimizes $\sum_{i=1}^n (Y_i - a - bX_i)^2$

Linear Regression (continued)

Finding $\hat{a}$ and $\hat{b}$.

- Let $S = S(a, b) = \sum_{i=1}^n (Y_i - a - bX_i)^2$.
- Minimize S .
- $\frac{\partial S}{\partial a} = -2 \sum_{i=1}^n (Y_i - a - bX_i)$ and $\frac{\partial S}{\partial b} = -2 \sum_{i=1}^n (Y_i - a - bX_i)X_i$.
- Solution is to find (a, b) such that $\frac{\partial S}{\partial a} = \frac{\partial S}{\partial b} = 0$
- The estimates are $\hat{a}$ of a and $\hat{b}$ of b .
- $\hat{a} = \bar{Y} - \hat{b}\bar{X}$
- $\hat{b} = \frac{s_{XY}}{s_{XX}}$
where $s_{XY} = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$, $s_{XX} = \sum_{i=1}^n (X_i - \bar{X})^2$.

Example 3 (continued)

- $\frac{\partial^2 S}{\partial a^2} = -2 \sum_{i=1}^n (-1) = 2n$, $\frac{\partial^2 S}{\partial b^2} = 2 \sum_{i=1}^n X_i^2$, $\frac{\partial^2 S}{\partial a \partial b} = 2 \sum_{i=1}^n X_i = 2n\bar{X}$
- The Hessian matrix

$$H^* = \begin{bmatrix} \frac{\partial^2 S}{\partial a^2} & \frac{\partial^2 S}{\partial a \partial b} \\ \frac{\partial^2 S}{\partial a \partial b} & \frac{\partial^2 S}{\partial b^2} \end{bmatrix}$$

i.e.,

$$H^* = \begin{bmatrix} n & n\bar{X} \\ n\bar{X} & \sum_{i=1}^n X_i^2 \end{bmatrix}$$

- $\det(H^*) = 4n \sum_{i=1}^n (X_i - \bar{X})^2 \geq 0$. The case when $\sum_{i=1}^n (X_i - \bar{X})^2 = 0$, all observations are identical and this case is not interesting. Therefore, hence $\det(H^*) > 0$ and S has a minimum at $(\hat{a}, \hat{b})$.
- Observe that as the $\hat{a}$ and $\hat{b}$ depend on the sample i.e., specific measurements out of all possible measurements, their values too are “probabilistic” i.e., they take certain values with certain probabilities.

Example

Dataset:

X_i	1	2	3	4	5
Y_i	2	3	5	4	6

Step 1: Regression model

We assume the simple linear regression model

$$Y = \hat{a} + \hat{b}X$$

where a is the intercept and b is the slope.

Step 2: Formulas

$$\hat{b} = \frac{\sum_{i=1}^5 (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^5 (X_i - \bar{X})^2}, \quad \hat{a} = \bar{Y} - \hat{b}\bar{X}$$

Step 3: Compute means

$$\bar{X} = \frac{1+2+3+4+5}{5} = 3, \quad \bar{Y} = \frac{2+3+5+4+6}{5} = 4$$

Exmaple (contined)

Step 4: Compute slope $\hat{b}$

X_i	Y_i	$(X_i - \bar{X})$	$(Y_i - \bar{Y})$	$(X_i - \bar{X})(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$
1	2	-2	-2	4	4
2	3	-1	-1	1	1
3	5	0	1	0	0
4	4	1	0	0	1
5	6	2	2	4	4
				Total = 9	Total = 10

$$\hat{b} = \frac{9}{10} = 0.9$$

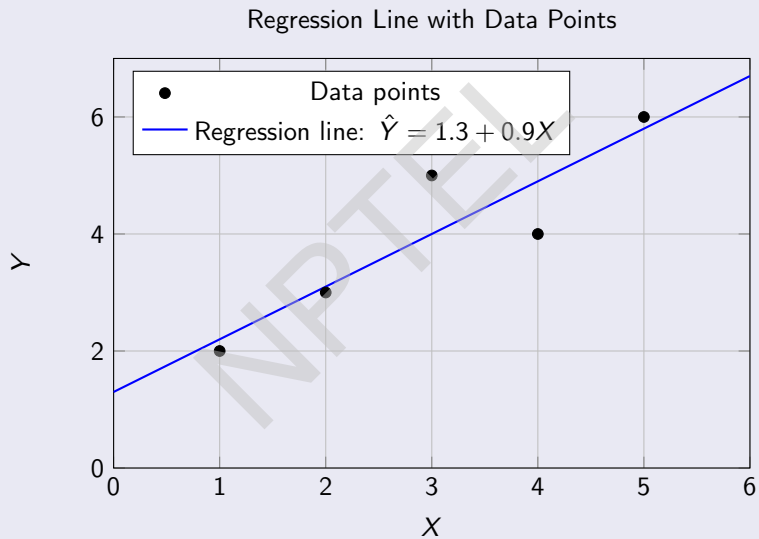
Step 5: Compute intercept $\hat{a}$

$$\hat{a} = \bar{Y} - \hat{b}\bar{X} = 4 - (0.9)(3) = 1.3$$

Step 6: Regression equation

$$\hat{Y} = 1.3 + 0.9X$$

Regression picture



1. Unbiasedness

Remember that $\hat{b} = \frac{s_{XY}}{s_{XX}}$ and $\hat{a} = \bar{Y} - \hat{b}\bar{X}$ are linear combinations of $Y_i, i = 1, 2, \dots, n$. Therefore

$$\hat{b} = \sum_{i=1}^n k_i Y_i$$

where $k_i = \frac{(X_i - \bar{X})}{s_{XX}}$ where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. Note that $\sum_{i=1}^n k_i = 0$ and $\sum_{i=1}^n k_i X_i = 1$. So,

$$\begin{aligned} E(\hat{b}) &= \sum_{i=1}^n k_i E(Y_i) \\ &= \sum_{i=1}^n k_i (a + bX_i) \\ &= b \end{aligned}$$

Therefore $\hat{b}$ is an unbiased estimator of b .

Unbiasedness (contined)

$$\begin{aligned} E(\hat{a}) &= E[\bar{Y} - \hat{b}\bar{X}] \\ &= E[a + b\bar{X} + \bar{\varepsilon} - \hat{b}\bar{X}] \\ &= a + b\bar{X} - b\bar{X} \\ &= a \end{aligned}$$

Thus $\hat{a}$ is an unbiased estimator of a .

2. Variances

We are assuming that Y_i 's are independently distributed,

$$\begin{aligned}\text{Var}(\hat{b}) &= \sum_{i=1}^n k_i^2 \text{Var}(Y_i) + \sum_i \sum_{j \neq i} k_i k_j \text{Cov}(Y_i, Y_j) \\ &= \sigma^2 \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{s_{XX}^2} \\ &= \frac{\sigma^2 s_{XX}}{s_{XX}^2} \\ &= \frac{\sigma^2}{s_{XX}}\end{aligned}$$

Here we have used the fact that $\text{Cov}(Y_i, Y_j) = 0$ as $Y_1, \dots, Y_n$ are independent.

Variances (continued)

Also

$$\text{Var}(\hat{a}) = \text{Var}(\bar{Y}) + \bar{X}^2 \text{Var}(\hat{b}) - 2\bar{X} \text{Cov}(\bar{Y}, \hat{b})$$

Now,

$$\begin{aligned}\text{Cov}(\bar{Y}, \hat{b}) &= E[\{\bar{Y} - E(\bar{Y})\}\{\hat{b} - E(\hat{b})\}] \\&= E\left[\bar{\varepsilon}\left(\sum_{i=1}^n k_i Y_i - b\right)\right] \\&= \frac{1}{n} E\left[\left(\sum_{i=1}^n \varepsilon_i\right)\left(a \sum_{i=1}^n k_i + b \sum_{i=1}^n c_i X_i + \sum_{i=1}^n k_i \varepsilon_i\right) - b \sum_{i=1}^n \varepsilon_i\right] \\&= \frac{1}{n} [0 + 0 + 0 + 0] \\&= 0\end{aligned}$$

So,

$$\text{Var}(\hat{a}) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{s_{XX}} \right)$$

3. Residual sum of squares

The residual sum of squares is given as

$$\begin{aligned}SS_{res} &= \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \\&= \sum_{i=1}^n (Y_i - \hat{a} - \hat{b}X_i)^2 \\&= \sum_{i=1}^n [Y_i - \bar{Y} + \hat{b}\bar{X} - \hat{b}X_i]^2 \\&= \sum_{i=1}^n [(Y_i - \bar{Y}) - \hat{b}(X_i - \bar{X})]^2\end{aligned}$$

Residual sum of squares (continued)

$$\begin{aligned}SS_{res} &= \sum_{i=1}^n (Y_i - \bar{Y})^2 + \hat{b}^2 \sum_{i=1}^n (X_i - \bar{X})^2 - 2\hat{b} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) \\&= s_{YY} + \hat{b}^2 s_{XX} - 2\hat{b} s_{XY} \\&= s_{YY} - \hat{b}^2 s_{XX} \\&= s_{YY} - \left(\frac{s_{XY}}{s_{XX}} \right)^2 s_{XX} \\&= s_{YY} - \frac{s_{XY}^2}{s_{XX}} \\&= s_{YY} - \hat{b} s_{XY}.\end{aligned}$$

where

$$s_{YY} = \sum_{i=1}^n (Y_i - \bar{Y})^2, \quad \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

4. Covariance

The covariance between $\hat{a}$ and $\hat{b}$ is

$$\text{Cov}(\hat{a}, \hat{b}) = \text{Cov}(\bar{Y}, \hat{b}) - \bar{X} \text{Var}(\hat{b}) = -\frac{\bar{X}}{s_{XX}} \sigma^2.$$

It can further be shown that the ordinary least squares estimators a and b possess the minimum variance in the class of linear and unbiased estimators. So they are termed as the Best Linear Unbiased Estimators (BLUE). Such a property is known as the Gauss–Markov theorem, which is discussed later in multiple linear regression model.

5. Estimation of σ^2

The estimator of σ^2 is obtained from the residual sum of squares as follows. Assuming that Y_i is normally distributed, it follows that SS_{res} has a χ^2 distribution with $(n - 2)$ degrees of freedom, so

$$\frac{SS_{res}}{\sigma^2} \sim \chi^2(n - 2).$$

Thus using the result about the expectation of a chi-square random variable, we have

$$E(SS_{res}) = (n - 2)\sigma^2.$$

Thus an unbiased estimator of σ^2 is

$$s^2 = \frac{SS_{res}}{n - 2}.$$

Note that SS_{res} has only $(n - 2)$ degrees of freedom. The two degrees of freedom are lost due to estimation of $\hat{a}$ and $\hat{b}$. Since s^2 depends on the estimates $\hat{a}$ and $\hat{b}$, so it is a **model-dependent estimate of σ^2** .

6. Estimates of $\hat{a}$ and $\hat{b}$

The estimators of variances of $\hat{a}$ and $\hat{b}$ are obtained by replacing σ^2 by its estimate $\hat{\sigma}^2 = s^2$ as follows:

$$\widehat{\text{Var}}(\hat{a}) = s^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{s_{XX}} \right)$$

and

$$\widehat{\text{Var}}(\hat{b}) = \frac{s^2}{s_{XX}}.$$

It is observed that since

$$\sum_{i=1}^n (Y_i - \hat{Y}_i) = 0, \quad \text{so} \quad \sum_{i=1}^n e_i = 0.$$

In the light of this property, e_i can be regarded as an estimate of the unknown ε_i ($i = 1, 2, \dots, n$). This helps in verifying the different model assumptions on the basis of the given sample (X_i, Y_i) , $i = 1, 2, \dots, n$.

Estimates of $\hat{a}$ and $\hat{b}$ (continued)

In the light of this property, e_i can be regarded as an estimate of the unknown ε_i ($i = 1, 2, \dots, n$). This helps in verifying the different model assumptions on the basis of the given sample (X_i, Y_i) , $i = 1, 2, \dots, n$. **Further, note that**

$$(i) \sum_{i=1}^n X_i e_i = 0,$$

$$(ii) \sum_{i=1}^n \hat{Y}_i e_i = 0,$$

$$(iii) \sum_{i=1}^n Y_i = \sum_{i=1}^n \hat{Y}_i,$$

(iv) the fitted line always passes through $(\bar{X}, \bar{Y})$.

Centered Model

Sometimes it is useful to measure the independent variable around its mean. In such a case, the model

$$Y_i = a + bX_i + \varepsilon_i$$

has a centred version as follows:

$$\begin{aligned} Y_i &= a + b(X_i - \bar{X}) + \varepsilon_i \quad (i = 1, 2, \dots, n) \\ &= a^* + b(X_i - \bar{X}) + \varepsilon_i \end{aligned}$$

where $a^* = a + b\bar{X}$. The sum of squares due to error is given by

$$S(a^*, b) = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n [Y_i - a^* - b(X_i - \bar{X})]^2.$$

Centered Model (continued)

Now solving

$$\frac{\partial S(a^*, b)}{\partial a^*} = 0$$

$$\frac{\partial S(a^*, b)}{\partial b} = 0,$$

we get the direct regression least squares estimates of a^* and b as

$$\hat{a}^* = \bar{Y}, \quad \hat{b} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{s_{XY}}{s_{XX}}.$$

respectively.

Centered Model (continued)

Thus the form of the estimate of slope parameter b remains the same in the usual and centered model whereas the form of the estimate of intercept term changes in the usual and centered models.

Further, the Hessian matrix of the second order partial derivatives of

$$S(a^*, b)$$

with respect to a^* and b is positive definite at $a^* = \hat{a}^*$ and $b = \hat{b}$, which ensures that $S(a^*, b)$ is minimized at $a^* = \hat{a}$ and $b = \hat{b}$.

Under the assumption that

$$E(\varepsilon_i) = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2, \quad \text{and} \quad \text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \quad \text{for all } i \neq j = 1, 2, \dots, n,$$

it follows that

$$\begin{aligned} E(\hat{a}^*) &= a^*, \quad E(\hat{b}) = b, \\ \text{Var}(\hat{a}^*) &= \frac{\sigma^2}{n}, \quad \text{Var}(\hat{b}) = \frac{\sigma^2}{S_{XX}}. \end{aligned}$$

Centered Model (continued)

In this case, the fitted model of

$$Y_i = a^* + b(X_i - \bar{X}) + \varepsilon_i$$

is

$$Y = \bar{Y} + \hat{b}(X - \bar{X}),$$

and the predicted values are

$$\hat{Y}_i = \bar{Y} + \hat{b}(X_i - \bar{X}) \quad (i = 1, \dots, n).$$

Note that in the centered model

$$\text{Cov}(\hat{a}^*, \hat{b}) = 0.$$

References

- 1 <https://home.iitk.ac.in/~shalab/regression/Chapter2-Regression-SimpleLinearRegressionAnalysis.pdf>
- 2 Introduction to Linear Regression Analysis by Douglas C. Montgomery, Elizabeth A. Peck, G. Geoffrey Vining (Wiley)
- 3 Applied Regression Analysis by Norman R. Draper, Harry Smith (Wiley)
- 4 <https://web.stanford.edu/class/stats202/notes/Linear-regression/Simple-linear-regression.html>

Thank you





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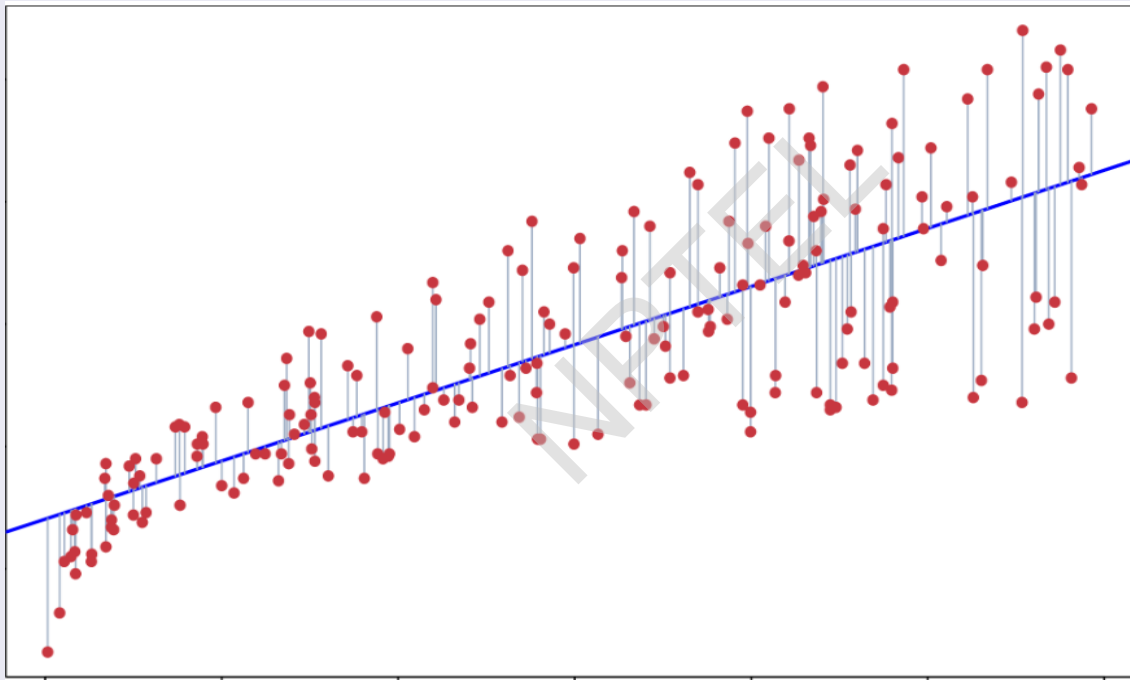
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A simple picture of regression



Linear Regression

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- One decent guess of f : the least square line !!
- Least Square Line: Line $Y = \hat{a} + \hat{b}X$ such that $a = \hat{a}$ and $b = \hat{b}$ minimizes $\sum_{i=1}^n (Y_i - a - bX_i)^2$

Linear Regression (continued)

Finding $\hat{a}$ and $\hat{b}$.

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- Minimize S .
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Example 3 (continued)

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$$H^* = \begin{bmatrix} \frac{\partial^2 S}{\partial a^2} & \frac{\partial^2 S}{\partial a \partial b} \\ \frac{\partial^2 S}{\partial a \partial b} & \frac{\partial^2 S}{\partial b^2} \end{bmatrix}$$

i.e.,

$$H^* = \begin{bmatrix} n & n\bar{X} \\ n\bar{X} & \sum_{i=1}^n X_i^2 \end{bmatrix}$$

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Example

Dataset:

X_i	1	2	3	4	5
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We assume the simple linear regression model

$$Y = \hat{a} + \hat{b}X$$

where a is the intercept and b is the slope.

Step 2: Formulas

$$\hat{b} = \frac{\sum_{i=1}^5 (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^5 (X_i - \bar{X})^2}, \quad \hat{a} = \bar{Y} - \hat{b}\bar{X}$$

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$$\bar{X} = \frac{1+2+3+4+5}{5} = 3, \quad \bar{Y} = \frac{2+3+5+4+6}{5} = 4$$

Exmaple (contined)

Step 4: Compute slope $\hat{b}$

X_i	Y_i	$(X_i - \bar{X})$	$(Y_i - \bar{Y})$	$(X_i - \bar{X})(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$
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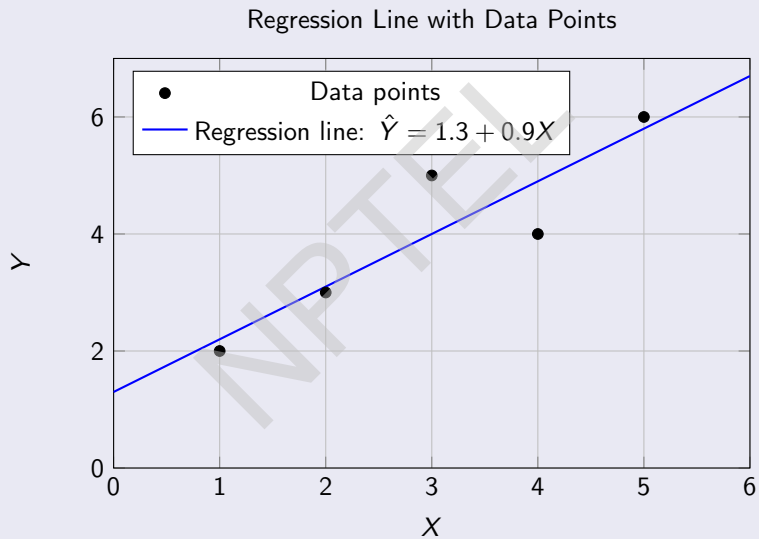
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$$\hat{a} = \bar{Y} - \hat{b}\bar{X} = 4 - (0.9)(3) = 1.3$$

Step 6: Regression equation

$$\hat{Y} = 1.3 + 0.9X$$

Regression picture



1. Unbiasedness

Remember that $\hat{b} = \frac{s_{XY}}{s_{XX}}$ and $\hat{a} = \bar{Y} - \hat{b}\bar{X}$ are linear combinations of $Y_i, i = 1, 2, \dots, n$. Therefore

$$\hat{b} = \sum_{i=1}^n k_i Y_i$$

where $k_i = \frac{(X_i - \bar{X})}{s_{XX}}$ where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. Note that $\sum_{i=1}^n k_i = 0$ and $\sum_{i=1}^n k_i X_i = 1$. So,

$$\begin{aligned} E(\hat{b}) &= \sum_{i=1}^n k_i E(Y_i) \\ &= \sum_{i=1}^n k_i (a + bX_i) \\ &= b \end{aligned}$$

Therefore $\hat{b}$ is an unbiased estimator of b .

Unbiasedness (contined)

$$\begin{aligned}E(\hat{a}) &= E[\bar{Y} - \hat{b}\bar{X}] \\&= E[a + b\bar{X} + \bar{\varepsilon} - \hat{b}\bar{X}] \\&= a + b\bar{X} - b\bar{X} \\&= a\end{aligned}$$

Thus $\hat{a}$ is an unbiased estimator of a .

2. Variances

We are assuming that Y_i 's are independently distributed,

$$\begin{aligned}\text{Var}(\hat{b}) &= \sum_{i=1}^n k_i^2 \text{Var}(Y_i) + \sum_i \sum_{j \neq i} k_i k_j \text{Cov}(Y_i, Y_j) \\ &= \sigma^2 \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{s_{XX}^2} \\ &= \frac{\sigma^2 s_{XX}}{s_{XX}^2} \\ &= \frac{\sigma^2}{s_{XX}}\end{aligned}$$

Here we have used the fact that $\text{Cov}(Y_i, Y_j) = 0$ as $Y_1, \dots, Y_n$ are independent.

Variances (continued)

Also

$$\text{Var}(\hat{a}) = \text{Var}(\bar{Y}) + \bar{X}^2 \text{Var}(\hat{b}) - 2\bar{X} \text{Cov}(\bar{Y}, \hat{b})$$

Now,

$$\begin{aligned}\text{Cov}(\bar{Y}, \hat{b}) &= E[\{\bar{Y} - E(\bar{Y})\}\{\hat{b} - E(\hat{b})\}] \\&= E\left[\bar{\varepsilon}\left(\sum_{i=1}^n k_i Y_i - b\right)\right] \\&= \frac{1}{n} E\left[\left(\sum_{i=1}^n \varepsilon_i\right)\left(a \sum_{i=1}^n k_i + b \sum_{i=1}^n c_i X_i + \sum_{i=1}^n k_i \varepsilon_i\right) - b \sum_{i=1}^n \varepsilon_i\right] \\&= \frac{1}{n} [0 + 0 + 0 + 0] \\&= 0\end{aligned}$$

So,

$$\text{Var}(\hat{a}) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{s_{XX}} \right)$$

3. Residual sum of squares

The residual sum of squares is given as

$$\begin{aligned}SS_{res} &= \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \\&= \sum_{i=1}^n (Y_i - \hat{a} - \hat{b}X_i)^2 \\&= \sum_{i=1}^n [Y_i - \bar{Y} + \hat{b}\bar{X} - \hat{b}X_i]^2 \\&= \sum_{i=1}^n [(Y_i - \bar{Y}) - \hat{b}(X_i - \bar{X})]^2\end{aligned}$$

Residual sum of squares (continued)

$$\begin{aligned}SS_{res} &= \sum_{i=1}^n (Y_i - \bar{Y})^2 + \hat{b}^2 \sum_{i=1}^n (X_i - \bar{X})^2 - 2\hat{b} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) \\&= s_{YY} + \hat{b}^2 s_{XX} - 2\hat{b} s_{XY} \\&= s_{YY} - \hat{b}^2 s_{XX} \\&= s_{YY} - \left(\frac{s_{XY}}{s_{XX}} \right)^2 s_{XX} \\&= s_{YY} - \frac{s_{XY}^2}{s_{XX}} \\&= s_{YY} - \hat{b} s_{XY}.\end{aligned}$$

where

$$s_{YY} = \sum_{i=1}^n (Y_i - \bar{Y})^2, \quad \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

4. Covariance

The covariance between $\hat{a}$ and $\hat{b}$ is

$$\text{Cov}(\hat{a}, \hat{b}) = \text{Cov}(\bar{Y}, \hat{b}) - \bar{X} \text{Var}(\hat{b}) = -\frac{\bar{X}}{s_{XX}} \sigma^2.$$

It can further be shown that the ordinary least squares estimators a and b possess the minimum variance in the class of linear and unbiased estimators. So they are termed as the Best Linear Unbiased Estimators (BLUE). Such a property is known as the Gauss–Markov theorem, which is discussed later in multiple linear regression model.

5. Estimation of σ^2

The estimator of σ^2 is obtained from the residual sum of squares as follows. Assuming that Y_i is normally distributed, it follows that SS_{res} has a χ^2 distribution with $(n - 2)$ degrees of freedom, so

$$\frac{SS_{res}}{\sigma^2} \sim \chi^2(n - 2).$$

Thus using the result about the expectation of a chi-square random variable, we have

$$E(SS_{res}) = (n - 2)\sigma^2.$$

Thus an unbiased estimator of σ^2 is

$$s^2 = \frac{SS_{res}}{n - 2}.$$

Note that SS_{res} has only $(n - 2)$ degrees of freedom. The two degrees of freedom are lost due to estimation of $\hat{a}$ and $\hat{b}$. Since s^2 depends on the estimates $\hat{a}$ and $\hat{b}$, so it is a **model-dependent estimate of σ^2** .

6. Estimates of $\hat{a}$ and $\hat{b}$

The estimators of variances of $\hat{a}$ and $\hat{b}$ are obtained by replacing σ^2 by its estimate $\hat{\sigma}^2 = s^2$ as follows:

$$\widehat{\text{Var}}(\hat{a}) = s^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{s_{XX}} \right)$$

and

$$\widehat{\text{Var}}(\hat{b}) = \frac{s^2}{s_{XX}}.$$

It is observed that since

$$\sum_{i=1}^n (Y_i - \hat{Y}_i) = 0, \quad \text{so} \quad \sum_{i=1}^n e_i = 0.$$

In the light of this property, e_i can be regarded as an estimate of the unknown ε_i ($i = 1, 2, \dots, n$). This helps in verifying the different model assumptions on the basis of the given sample (X_i, Y_i) , $i = 1, 2, \dots, n$.

Estimates of $\hat{a}$ and $\hat{b}$ (continued)

In the light of this property, e_i can be regarded as an estimate of the unknown ε_i ($i = 1, 2, \dots, n$). This helps in verifying the different model assumptions on the basis of the given sample (X_i, Y_i) , $i = 1, 2, \dots, n$. **Further, note that**

$$(i) \sum_{i=1}^n X_i e_i = 0,$$

$$(ii) \sum_{i=1}^n \hat{Y}_i e_i = 0,$$

$$(iii) \sum_{i=1}^n Y_i = \sum_{i=1}^n \hat{Y}_i,$$

(iv) the fitted line always passes through $(\bar{X}, \bar{Y})$.

Centered Model

Sometimes it is useful to measure the independent variable around its mean. In such a case, the model

$$Y_i = a + bX_i + \varepsilon_i$$

has a centred version as follows:

$$\begin{aligned} Y_i &= a + b(X_i - \bar{X}) + \varepsilon_i \quad (i = 1, 2, \dots, n) \\ &= a^* + b(X_i - \bar{X}) + \varepsilon_i \end{aligned}$$

where $a^* = a + b\bar{X}$. The sum of squares due to error is given by

$$S(a^*, b) = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n [Y_i - a^* - b(X_i - \bar{X})]^2.$$

Centered Model (continued)

Now solving

$$\frac{\partial S(a^*, b)}{\partial a^*} = 0$$

$$\frac{\partial S(a^*, b)}{\partial b} = 0,$$

we get the direct regression least squares estimates of a^* and b as

$$\hat{a}^* = \bar{Y}, \quad \hat{b} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{s_{XY}}{s_{XX}}.$$

respectively.

Centered Model (continued)

Thus the form of the estimate of slope parameter b remains the same in the usual and centered model whereas the form of the estimate of intercept term changes in the usual and centered models.

Further, the Hessian matrix of the second order partial derivatives of

$$S(a^*, b)$$

with respect to a^* and b is positive definite at $a^* = \hat{a}^*$ and $b = \hat{b}$, which ensures that $S(a^*, b)$ is minimized at $a^* = \hat{a}$ and $b = \hat{b}$.

Under the assumption that

$$E(\varepsilon_i) = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2, \quad \text{and} \quad \text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \quad \text{for all } i \neq j = 1, 2, \dots, n,$$

it follows that

$$\begin{aligned} E(\hat{a}^*) &= a^*, \quad E(\hat{b}) = b, \\ \text{Var}(\hat{a}^*) &= \frac{\sigma^2}{n}, \quad \text{Var}(\hat{b}) = \frac{\sigma^2}{s_{xx}}. \end{aligned}$$

Centered Model (continued)

In this case, the fitted model of

$$Y_i = a^* + b(X_i - \bar{X}) + \varepsilon_i$$

is

$$Y = \bar{Y} + \hat{b}(X - \bar{X}),$$

and the predicted values are

$$\hat{Y}_i = \bar{Y} + \hat{b}(X_i - \bar{X}) \quad (i = 1, \dots, n).$$

Note that in the centered model

$$\text{Cov}(\hat{a}^*, \hat{b}) = 0.$$

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- 2 Introduction to Linear Regression Analysis by Douglas C. Montgomery, Elizabeth A. Peck, G. Geoffrey Vining (Wiley)
- 3 Applied Regression Analysis by Norman R. Draper, Harry Smith (Wiley)

Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

PROF. ARINDAM BANERJEE

Department of Mathematics
INDIAN INSTITUTE OF TECHNOLOGY KHARGPUR

Module 06: Regression

Lecture 03: Multiple Linear Regression 1



Linear Modelling

Setup:

- 1 Independent variables: $X_1, \dots, X_k$.
- 2 Dependent variable: Y .
- 3 $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \varepsilon$ where $E(\varepsilon) = 0$ and $\text{Var}(\varepsilon) = \sigma^2$

Goal: To model the above using i.i.d. sample of n data points from the above population $(X_{11}, \dots, X_{1k}, Y_1), \dots, (X_{n1}, \dots, X_{nk}, Y_n)$. So, for all i ,

$$Y_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_k X_{ik} + \varepsilon_i$$

where $E(\varepsilon_i) = 0$ and $\text{Var}(\varepsilon_i) = \sigma^2$.

Specifically speaking we want to guess β_i s from the data.

1. Linear Regression Model (Scalar Form)

We assume a linear relationship between the dependent variable Y and independent variables $X_1, X_2, \dots, X_k$:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n.$$

where:

- Y_i = i -th sample point for dependent variable.
- X_{ij} = i -th sample point for j -th independent variable.
- β_j = regression coefficients,
- ε_i = random error term for i -th sample point.

2. Matrix Representation

The same model in matrix notation:

$$\underline{Y} = X\beta + \varepsilon,$$

i.e,

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & X_{12} & \cdots & X_{1k} \\ 1 & X_{21} & X_{22} & \cdots & X_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{nk} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

where

$$\underline{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}_{n \times 1}, \quad X = \begin{bmatrix} 1 & X_{11} & X_{12} & \cdots & X_{1k} \\ 1 & X_{21} & X_{22} & \cdots & X_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{nk} \end{bmatrix},$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

3. Objective of OLS

The goal of OLS is to find β that minimizes the total squared error:

$$S(\beta) = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = (\underline{Y} - X\beta)^\top (\underline{Y} - X\beta) = \|\underline{Y} - X\beta\|^2 = \|\varepsilon\|^2$$
$$\Rightarrow \min_{\beta} S(\beta)$$

Hence,

$$\hat{\beta}_{OLS} = \arg \min_{\beta} (\underline{Y} - X\beta)^\top (\underline{Y} - X\beta)$$

4. Why It Is Called Ordinary Least Squares (OLS)

- **Ordinary:** All observations are given equal weight.
- **Least Squares:** The estimator minimizes the sum of squared residuals.

Residual for i^{th} observation: $e_i = Y_i - \hat{Y}_i$.

$$\text{OLS criterion: } \min \sum_{i=1}^n e_i^2.$$

where $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{i1} + \cdots + \hat{\beta}_k X_{ik}$

5. Derivation of the Normal Equations

Differentiating $S(\beta)$ with respect to β :

$$\frac{\partial S(\beta)}{\partial \beta} = -2X^T(\underline{Y} - X\beta)$$

Setting this derivative equal to zero gives:

$$X^T X \hat{\beta} = X^T \underline{Y}$$

Thus,

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T \underline{Y}$$

provided $X^T X$ is invertible.

Hessian Matrix

Here $S(\beta) = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_{i1} - \dots - \beta_k X_{ik})^2$.

The Hessian matrix is

$$2 \begin{bmatrix} \sum_{i=1}^n X_{i1} & \sum_{i=1}^n X_{i1}^2 & \dots & \sum_{i=1}^n X_{i1} X_{ik} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^n X_{ik} & \sum_{i=1}^n X_{ik} X_{i1} & \dots & \sum_{i=1}^n X_{ik}^2 \end{bmatrix} = 2X^T X$$

which is positive definite.

7. Key Insight

- OLS finds the coefficient vector $\hat{\beta}$ that minimizes squared prediction errors.
- The positive-definite Hessian ensures a unique minimum.
- This is the foundation on which the Gauss–Markov theorem is built.

Example: Ordinary Least Squares (OLS) Calculation

Given

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad \underline{Y} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}.$$

We want to find

$$\hat{\beta} = (X^T X)^{-1} X^T \underline{Y}.$$

Computation

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \quad X^T \underline{Y} = \begin{bmatrix} 10 \\ 23 \end{bmatrix}.$$

The inverse is

$$(X^T X)^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}.$$

Now multiply:

$$\begin{aligned} \hat{\beta} &= (X^T X)^{-1} X^T \underline{Y} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 10 \\ 23 \end{bmatrix} \\ \Rightarrow \hat{\beta} &= \frac{1}{6} \begin{bmatrix} 2 \\ 9 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{3}{2} \end{bmatrix}. \end{aligned}$$

Result

The estimated regression coefficients are:

$$\hat{\beta}_0 = 0.333, \quad \hat{\beta}_1 = 1.5$$

Hence the fitted regression line is:

$$\hat{Y} = 0.333 + 1.5X.$$

References

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- 2 <https://mathvoices.ams.org/featurecolumn/2022/03/01/ordinary-least-squares/>
- 3 Wikipedia

Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

PROF. ARINDAM BANERJEE

Department of Mathematics
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Module 06: Regression

Lecture 04: Multiple Linear Regression 2



Linear Modelling

Setup:

- 1 Independent variables: $X_1, \dots, X_k$.
- 2 Dependent variable: Y .
- 3 $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \varepsilon$ where $E(\varepsilon) = 0$ and $\text{Var}(\varepsilon) = \sigma^2$

Goal: To model the above using i.i.d. sample of n data points from the above population $(X_{11}, \dots, X_{1k}, Y_1), \dots, (X_{n1}, \dots, X_{nk}, Y_n)$. So, for all i ,

$$Y_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_k X_{ik} + \varepsilon_i$$

where $E(\varepsilon_i) = 0$ and $\text{Var}(\varepsilon_i) = \sigma^2$.

Specifically speaking, we want to guess β_i s from the data.

Multiple Linear Regression Model

We have a linear relationship between the dependent variable y and independent variables $x_1, x_2, \dots, x_k$:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n.$$

where:

- Y_i = i -th sample point for dependent variable.
- X_{ij} = i -th sample point for j -th independent variable.
- β_j = regression coefficients,
- ε_i = random error term for i -th sample point.

Multiple Linear Regression Model (continued)

- The same model in matrix notation:

$$\underline{Y} = X\beta + \varepsilon,$$

where

$$\underline{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}_{n \times 1}, \quad X = \begin{bmatrix} 1 & X_{11} & X_{12} & \cdots & X_{1k} \\ 1 & X_{21} & X_{22} & \cdots & X_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{nk} \end{bmatrix},$$
$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

- Remember that $(X^T X)^{-1} X^T \underline{Y}$ is the ordinary least square of β .

Multiple Linear Regression Model (continued)

Definitions:

- An $n \times n$ real symmetric matrix M is called a positive semi-definite if $x^T M x \geq 0$ for all $x \in \mathbb{R}^n$.
- For a random vector $Z = (Z_1, \dots, Z_n)$, $\text{Var}(Z)$ is the $n \times n$ matrix whose (i, j) -th entry is $\text{Cov}(Z_i, Z_j)$

Multiple Linear Regression Model (continued)

The best linear unbiased estimator of the vector β of parameters β_j , is one with the smallest mean squared error for every vector λ of linear combination parameters. This is equivalent to the condition that

$$\text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta})$$

is a positive semi-definite matrix for every other linear unbiased estimator $\tilde{\beta}$.

Assumptions

- $E(\varepsilon_i) = 0$.
- $\text{Var}(\varepsilon_i^2) = 0$.
- $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ as the drawn samples are i.i.d.

Gauss Markov Theorem

Consider the model

$$\underline{Y} = X\beta + \varepsilon,$$

where

$$\underline{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & X_{11} & X_{12} & \cdots & X_{1k} \\ 1 & X_{21} & X_{22} & \cdots & X_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{nk} \end{bmatrix},$$
$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

Then the ordinary least squares estimator $\hat{\beta}$ is the best linear unbiased estimator of β .

Proof of the Gauss-Markov Theorem

We will show $\hat{\beta}$ is an unbiased estimator of β where

$$\hat{\beta} = (X^T X)^{-1} X^T \underline{Y}$$

Let $Z = (X^T X)^{-1} X^T$, then $\hat{\beta} = Z \underline{Y}$.

Now,

$$\begin{aligned} E(\hat{\beta}) &= E(Z \underline{Y}) \\ &= E_{X, \underline{Y}}(Z \underline{Y}) \\ &= E_X(E_{\underline{Y}|X}(Z \underline{Y})) \\ &= E_X\left(Z E_{\underline{Y}|X} \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}\right) \end{aligned}$$

where

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik} + \varepsilon_i$$

and $E(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$

Proof of the Gauss-Markov Theorem (continued)

$$\begin{aligned} E(\hat{\beta}) &= E_X \left(Z \begin{bmatrix} E_{Y|X}(Y_1) \\ E_{Y|X}(Y_2) \\ \vdots \\ E_{Y|X}(Y_n) \end{bmatrix} \right) \\ &= E_X \left(Z \begin{bmatrix} \beta_0 + \beta_1 X_{11} + \cdots + \beta_k X_{1k} \\ \beta_0 + \beta_1 X_{21} + \cdots + \beta_k X_{2k} \\ \vdots \\ \beta_0 + \beta_1 X_{n1} + \cdots + \beta_k X_{nk} \end{bmatrix} \right) \\ &= E_X (ZX\beta) \\ &= \beta E_X (ZX) \\ &= \beta E_X \{ (X^\top X)^{-1} X^\top X \} \\ &= \beta \end{aligned}$$

Proof of the Gauss-Markov Theorem (continued)

Let $\tilde{\beta} = C\underline{Y}$ be another linear estimator of β with $C = (X^\top X)^{-1}X^\top + D$, where D is a $k \times n$ non-zero matrix. As we're restricting to unbiased estimators, minimum mean squared error implies minimum variance. The goal is therefore to show that such an estimator has a variance no smaller than that of $\hat{\beta}$, the OLS estimator.

Proof of the Gauss-Markov Theorem (continued)

We calculate,

$$\begin{aligned} E(\tilde{\beta}) &= E[C\underline{Y}] \\ &= E[((X^\top X)^{-1}X^\top + D)(X\beta + \varepsilon)] \\ &= ((X^\top X)^{-1}X^\top + D)X\beta + ((X^\top X)^{-1}X^\top + D)E[\varepsilon] \\ &= ((X^\top X)^{-1}X^\top + D)X\beta \\ &= ((X^\top X)^{-1}X^\top)X\beta + DX\beta \\ &= (I_k + DX)\beta \end{aligned}$$

Since β is unobservable, $\tilde{\beta}$, is unbiased iff $DX = 0$.

Proof of the Gauss-Markov Theorem (continued)

Now

$$\begin{aligned}\text{Var}(\tilde{\beta}) &= \text{Var}(CY) \\ &= C\text{Var}(\underline{Y})C^\top \\ &= \sigma^2 CC^\top \\ &= \sigma^2((X^\top X)^{-1}X^\top + D)(X(X^\top X)^{-1} + D^\top) \\ &= \sigma^2((X^\top X)^{-1}X^\top X(X^\top X)^{-1} + (X^\top X)^{-1}X^\top D^\top + DX(X^\top X)^{-1} + DD^\top) \\ &= \sigma^2(X^\top X)^{-1} + \sigma^2(X^\top X)^{-1}(DX)^\top + \sigma^2 DX(X^\top X)^{-1} + \sigma^2 DD^\top \\ &= \sigma^2(X^\top X)^{-1} + \sigma^2 DD^\top \quad [DX = 0] \\ &= \text{Var}(\hat{\beta}) + \sigma^2 DD^\top \quad [\sigma^2(X^\top X)^{-1} = \text{Var}(\hat{\beta})]\end{aligned}$$

Since $DD^\top$ is a positive semidefinite matrix, $\text{Var}(\tilde{\beta})$ exceeds by $\text{Var}(\hat{\beta})$ by a positive semidefinite matrix.

Maximum Likelihood Estimator of Ordinary Least Squares

We assume that ε_i 's ($i = 1, 2, \dots, n$) are independent and identically distributed following a normal distribution

$$\mathcal{N}(0, \sigma^2).$$

Now we use the method of maximum likelihood to estimate the parameters of the linear regression model

$$Y_i = a + bX_i + \varepsilon_i, \quad (i = 1, 2, \dots, n),$$

the observations Y_i ($i = 1, 2, \dots, n$) are independently distributed with

$$\mathcal{N}(a + bX_i, \sigma^2)$$

for all $i = 1, 2, \dots, n$.

Maximum Likelihood Estimator of Simple Linear Regression (continued)

The likelihood function of the given observations (X_i, Y_i) and unknown parameters a, b and σ^2 is

$$L(X_i, Y_i; a, b, \sigma^2) = \prod_{i=1}^n \left(\frac{1}{2\pi\sigma^2} \right)^{1/2} \exp \left[-\frac{1}{2\sigma^2} (Y_i - a - bX_i)^2 \right].$$

The maximum likelihood estimates of a, b and σ^2 can be obtained by maximizing $L(X_i, Y_i; a, b, \sigma^2)$ or equivalently its log-likelihood function $\ln L(X_i, Y_i; a, b, \sigma^2)$, where

$$\ln L(X_i, Y_i; a, b, \sigma^2) = -\left(\frac{n}{2}\right) \ln(2\pi) - \left(\frac{n}{2}\right) \ln(\sigma^2) - \left(\frac{1}{2\sigma^2}\right) \sum_{i=1}^n (Y_i - a - bX_i)^2.$$

Maximum Likelihood Estimator of Simple Linear Regression (continued)

The normal equations are obtained by partial differentiation of the log-likelihood with respect to a , b , and σ^2 , and equating them to zero as follows:

$$\frac{\partial \ln L(X_i, Y_i; a, b, \sigma^2)}{\partial a} = -\frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - a - bX_i) = 0$$

$$\frac{\partial \ln L(X_i, Y_i; a, b, \sigma^2)}{\partial b} = -\frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - a - bX_i)X_i = 0$$

and

$$\frac{\partial \ln L(X_i, Y_i; a, b, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (Y_i - a - bX_i)^2 = 0.$$

Maximum Likelihood Estimator of Simple Linear Regression (continued)

The solution of these normal equations gives the maximum likelihood estimates of a , b , and σ^2 as

$$\hat{a} = \bar{Y} - \hat{b}\bar{X},$$
$$\hat{b} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{s_{XY}}{s_{XX}},$$

and

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (Y_i - \hat{a} - \hat{b}X_i)^2}{n},$$

respectively. It can be verified that the Hessian matrix of second-order partial derivatives ensures that the likelihood function is maximized at these values.

Maximum likelihood estimation for multiple linear regression

Theorem: Given a *linear regression model* with correlated observations

$$\underline{Y} = X\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_n), \quad (1)$$

the *maximum likelihood estimates* of β and σ^2 are given by

$$\hat{\beta} = (X^\top X)^{-1} X^\top \underline{Y}$$

and

$$\hat{\sigma}^2 = \frac{1}{n} (\underline{Y} - X\hat{\beta})^\top (\underline{Y} - X\hat{\beta}). \quad (2)$$

where I_n is the identity matrix of order n .

Maximum likelihood estimation for multiple linear regression (continued)

Proof: With the *probability density function* of the multivariate normal distribution, the linear regression equation (1) implies the following likelihood function

$$\begin{aligned} p(\underline{Y} \mid \beta, \sigma^2) &= \mathcal{N}(\underline{Y}; \underline{X}\beta, \sigma^2 I_n) \\ &= \frac{1}{(2\pi)^{\frac{n}{2}} \sqrt{|\sigma^2 I_n|}} \exp \left[-\frac{1}{2} (\underline{Y} - \underline{X}\beta)^\top (\sigma^2 I_n)^{-1} (\underline{Y} - \underline{X}\beta) \right] \end{aligned} \quad (3)$$

Here

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

Using $|\sigma^2 I_n| = (\sigma^2)^n |I_n|$, the log-likelihood function

$$\begin{aligned} l(\beta, \sigma^2) &= \log p(\underline{Y} \mid \beta, \sigma^2) \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2} \log |I_n| - \frac{1}{2\sigma^2} (\underline{Y} - \underline{X}\beta)^\top I_n^{-1} (\underline{Y} - \underline{X}\beta). \end{aligned} \quad (4)$$

Maximum likelihood estimation for multiple linear regression (continued)

Substituting the precision matrix $P = I_n^{-1} = I_n$ into (4) to ease notation, we have:

$$l(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2} \log(|I_n|) - \frac{1}{2\sigma^2} (\underline{Y}^\top P \underline{Y} - 2\beta^\top X^\top P \underline{Y} + \beta^\top X^\top P X \beta). \quad (5)$$

The derivative of the log-likelihood function (5) with respect to β is

$$\begin{aligned} \frac{dl(\beta, \sigma^2)}{d\beta} &= \frac{d}{d\beta} \left(-\frac{1}{2\sigma^2} (\underline{Y}^\top P \underline{Y} - 2\beta^\top X^\top P \underline{Y} + \beta^\top X^\top P X \beta) \right) \\ &= \frac{1}{2\sigma^2} \frac{d}{d\beta} (2\beta^\top X^\top P \underline{Y} - \beta^\top X^\top P X \beta) \\ &= \frac{1}{2\sigma^2} (2X^\top P \underline{Y} - 2X^\top P X \beta) \\ &= \frac{1}{\sigma^2} (X^\top P \underline{Y} - X^\top P X \beta). \end{aligned} \quad (6)$$

Maximum likelihood estimation for multiple linear regression (continued)

and setting this derivative to zero gives the MLE for β :

$$\frac{dl(\beta, \sigma^2)}{d\beta} = 0$$

$$0 = \frac{1}{\sigma^2} (X^\top P \underline{Y} - X^\top P X \beta)$$

$$0 = X^\top P \underline{Y} - X^\top P X \beta$$

$$X^\top P X \beta = X^\top P \underline{Y}$$

i.e.,

$$\hat{\beta} = (X^\top P X)^{-1} X^\top P \underline{Y} \quad (7)$$

Maximum likelihood estimation for multiple linear regression (continued)

The derivative of the log-likelihood function (4) at $\hat{\beta}$ with respect to σ^2 is

$$\begin{aligned}\frac{dl(\beta, \sigma^2)}{d\sigma^2} &= \frac{d}{d\sigma^2} \left(-\frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} (\underline{Y} - X\hat{\beta})^\top (\underline{Y} - X\hat{\beta}) \right) \\ &= -\frac{n}{2} \cdot \frac{1}{\sigma^2} + \frac{1}{2(\sigma^2)^2} (\underline{Y} - X\hat{\beta})^\top (\underline{Y} - X\hat{\beta}) \\ &= -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} (\underline{Y} - X\hat{\beta})^\top (\underline{Y} - X\hat{\beta})\end{aligned}\tag{8}$$

Maximum likelihood estimation for multiple linear regression (continued)

and setting this derivative to zero gives the MLE for σ^2 :

$$\begin{aligned}\frac{dl(\beta, \sigma^2)}{d\sigma^2} &= 0 \\ 0 &= -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}(\underline{Y} - X\hat{\beta})^\top(\underline{Y} - X\hat{\beta}) \\ \frac{n}{2\sigma^2} &= \frac{1}{2(\sigma^2)^2}(\underline{Y} - X\hat{\beta})^\top(\underline{Y} - X\hat{\beta}) \\ 2(\sigma^2)^2 \cdot \frac{n}{2\sigma^2} &= 2(\sigma^2)^2 \cdot \frac{1}{2(\sigma^2)^2}(\underline{Y} - X\hat{\beta})^\top(\underline{Y} - X\hat{\beta})\end{aligned}$$

i.e.,

$$\hat{\sigma}^2 = \frac{1}{n}(\underline{Y} - X\hat{\beta})^\top(\underline{Y} - X\hat{\beta}) \quad (9)$$

Together, (7) and (9) constitute the MLE for multiple linear regression.

Testing of hypotheses for slope parameter

Case 1: When σ^2 is known

Consider the simple linear regression model

$$Y_i = a + bX_i + \varepsilon_i \quad (i = 1, 2, \dots, n).$$

It is assumed that the ε_i 's are independent and identically distributed and follow $\mathcal{N}(0, \sigma^2)$.

First, we develop a test for the null hypothesis related to the slope parameter

$$H_0 : b = 0$$

Testing of hypotheses for slope parameter (continued)

Assuming σ^2 to be known, we know that

$$E(\hat{b}) = b, \quad \text{Var}(\hat{b}) = \frac{\sigma^2}{s_{XX}}$$

where $s_{XX} = \sum_{i=1}^n (X_i - \bar{X})^2$ and $\hat{b}$ is a linear combination of normally distributed Y_i 's. So

$$\hat{b} \sim \mathcal{N}\left(b, \frac{\sigma^2}{s_{XX}}\right)$$

and so the following statistic can be constructed

$$Z_1 = \frac{\hat{b} - b}{\sqrt{\frac{\sigma^2}{s_{XX}}}}$$

which is distributed as $\mathcal{N}(0, 1)$ when H_0 is true.

Testing of hypotheses for slope parameter (continued)

A decision rule to test

$$H_1 : b \neq 0$$

can be framed as follows:

$$\text{Reject } H_0 \text{ if } |Z_1| > Z_{\alpha/2}$$

where $Z_{\alpha/2}$ is the $\alpha/2$ percent point on the normal distribution.

Similarly, the decision rule for one-sided alternative hypothesis can also be framed.

Testing of hypotheses for slope parameter (continued)

Case 2: When σ^2 is unknown:

When σ^2 is unknown then we proceed as follows. We know that

$$\frac{SS_{\text{res}}}{\sigma^2} \sim \chi^2(n-2)$$

and

$$E\left(\frac{SS_{\text{res}}}{n-2}\right) = \sigma^2.$$

where $SS_{\text{res}} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$

Testing of hypotheses slope parameter (continued)

Further, $\frac{SS_{\text{res}}}{\sigma^2}$ and b_1 are independently distributed. This result will be proved formally later in the next module on multiple linear regression.

This result also follows from the fact that under normal distribution, the maximum likelihood estimates, viz., the sample mean (estimator of population mean) and the sample variance (estimator of population variance) are independently distributed, so $\hat{b}$ and s^2 are also independently distributed.

Thus the following statistic can be constructed:

$$t_0 = \frac{\hat{b} - b}{\sqrt{\frac{\hat{\sigma}^2}{s_{XX}}}} = \frac{\hat{b} - b}{\sqrt{\frac{SS_{\text{res}}}{(n-2)s_{XX}}}}$$

which follows a t -distribution with $(n-2)$ degrees of freedom, denoted as t_{n-2} , when H_0 is true, here $s_{XX} = \sum_{i=1}^n (X_i - \bar{X})^2$.

Testing of hypotheses for intercept term

Case 1: When σ^2 is known

Suppose the null hypothesis under consideration is

$$H_0 : a = a_0,$$

where σ^2 is known. Then, using the result that

$$E(\hat{a}) = a, \quad \text{Var}(\hat{a}) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{s_{XX}} \right)$$

and a is a linear combination of normally distributed random variables, the following statistic

$$Z_0 = \frac{\hat{a} - a_0}{\sqrt{\sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{s_{XX}} \right)}}$$

has a $\mathcal{N}(0, 1)$ distribution when H_0 is true.

Testing of hypotheses for intercept term (continued)

A decision rule to test

$$H_1 : a \neq a_0$$

can be framed as follows:

$$\text{Reject } H_0 \text{ if } |Z_0| > Z_{\alpha/2}$$

where $Z_{\alpha/2}$ is the $\alpha/2$ percentage point on the normal distribution. Similarly, the decision rule for one-sided alternative hypothesis can also be framed.

Testing of hypotheses for intercept term (continued)

Case 2: When σ^2 is unknown

When σ^2 is unknown, then the following statistic is constructed

$$t_0 = \frac{\hat{a} - a_0}{\sqrt{\frac{SS_{\text{res}}}{n-2} \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{XX}} \right)}}$$

which follows a t -distribution with $(n-2)$ degrees of freedom, i.e., t_{n-2} , when H_0 is true.

A decision rule to test

$$H_1 : a \neq a_0$$

is as follows:

Reject H_0 whenever $|t_0| > t_{n-2, \alpha/2}$

where $t_{n-2, \alpha/2}$ is the $\alpha/2$ percentage point of the t -distribution with $(n-2)$ degrees of freedom. Similarly, the decision rule for one-sided alternative hypothesis can also be framed.

General Linear Hypothesis

Assume

$$Y \sim N(X\beta, \sigma^2 I_n),$$

where Y is $n \times 1$, X is $n \times p$, and β is $p \times 1$. Let $\text{rank}(X) = r$.

Then

$$RSS = \min_{\beta} (Y - X\beta)^{\top} (Y - X\beta),$$

and the minimum is attained at

$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} Y.$$

$$\frac{RSS}{\sigma^2} \sim \chi_{n-r}^2.$$

Law of Iterated Expectation (Vector Form)

For any integrable function $g : \mathbb{R}^{2n} \rightarrow \mathbb{R}$,

$$\mathbb{E}_{\mathbf{X}, \mathbf{Y}}[g(\mathbf{X}, \mathbf{Y})] = \mathbb{E}_{\mathbf{X}}[\mathbb{E}_{\mathbf{Y}|\mathbf{X}}[g(\mathbf{X}, \mathbf{Y})]].$$

Variances

We are assuming that Y_i 's are independently distributed,

$$\begin{aligned}\text{Var}(\hat{b}) &= \sum_{i=1}^n k_i^2 \text{Var}(Y_i) + \sum_i \sum_{j \neq i} k_i k_j \text{Cov}(Y_i, Y_j) \\ &= \sigma^2 \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{s_{XX}^2} \\ &= \frac{\sigma^2 s_{XX}}{s_{XX}^2} \\ &= \frac{\sigma^2}{s_{XX}}\end{aligned}$$

Here we have used the fact that $\text{Cov}(Y_i, Y_j) = 0$ as $Y_1, \dots, Y_n$ are independent.

Variances (continued)

Also

$$\text{Var}(\hat{a}) = \text{Var}(\bar{Y}) + \bar{X}^2 \text{Var}(\hat{b}) - 2\bar{X} \text{Cov}(\bar{Y}, \hat{b})$$

Now,

$$\begin{aligned}\text{Cov}(\bar{Y}, \hat{b}) &= E[\{\bar{Y} - E(\bar{Y})\}\{\hat{b} - E(\hat{b})\}] \\&= E\left[\bar{\varepsilon}\left(\sum_{i=1}^n k_i Y_i - b\right)\right] \\&= \frac{1}{n} E\left[\left(\sum_{i=1}^n \varepsilon_i\right)\left(a \sum_{i=1}^n k_i + b \sum_{i=1}^n c_i X_i + \sum_{i=1}^n k_i \varepsilon_i\right) - b \sum_{i=1}^n \varepsilon_i\right] \\&= \frac{1}{n} [0 + 0 + 0 + 0] \\&= 0\end{aligned}$$

So,

$$\mathbf{Var}(\hat{a}) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{s_{XX}} \right)$$

References

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Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

PROF. ARINDAM BANERJEE

Department of Mathematics
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Module 06: Regression

Lecture 05: Multiple Linear Regression 3



Linear Modelling

Setup:

- 1 Independent variables: $X_1, \dots, X_k$.
- 2 Dependent variable: Y .
- 3 $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \varepsilon$ where $E(\varepsilon) = 0$ and $\text{Var}(\varepsilon) = \sigma^2$

Goal: To model the above using i.i.d. sample of n data points from the above population $(X_{11}, \dots, X_{1k}, Y_1), \dots, (X_{n1}, \dots, X_{nk}, Y_n)$. So, for all i ,

$$Y_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_k X_{ik} + \varepsilon_i$$

where $E(\varepsilon_i) = 0$ and $\text{Var}(\varepsilon_i) = \sigma^2$.

Specifically speaking, we want to guess β_i s from the data.

Multiple Linear Regression Model

We have a linear relationship between the dependent variable y and independent variables $x_1, x_2, \dots, x_k$:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n.$$

where:

- Y_i = i -th sample point for dependent variable.
- X_{ij} = i -th sample point for j -th independent variable.
- β_j = regression coefficients,
- ε_i = random error term for i -th sample point.

Multiple Linear Regression Model (continued)

- The same model in matrix notation:

$$\underline{Y} = X\beta + \varepsilon,$$

where

$$\underline{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}_{n \times 1}, \quad X = \begin{bmatrix} 1 & X_{11} & X_{12} & \cdots & X_{1k} \\ 1 & X_{21} & X_{22} & \cdots & X_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{nk} \end{bmatrix},$$
$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

- Remember that $(X^\top X)^{-1} X^\top \underline{Y}$ is the ordinary least square of β .

Multiple Linear Regression Model (continued)

Definitions:

- An $n \times n$ real symmetric matrix M is called a positive semi-definite if $x^T M x \geq 0$ for all $x \in \mathbb{R}^n$.
- For a random vector $Z = (Z_1, \dots, Z_n)$, $\text{Var}(Z)$ is the $n \times n$ matrix whose (i, j) -th entry is $\text{Cov}(Z_i, Z_j)$

Multiple Linear Regression Model (continued)

The best linear unbiased estimator of the vector β of parameters β_j , is one with the smallest mean squared error for every vector λ of linear combination parameters. This is equivalent to the condition that

$$\text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta})$$

is a positive semi-definite matrix for every other linear unbiased estimator $\tilde{\beta}$.

Assumptions

- $E(\varepsilon_i) = 0$.
- $\text{Var}(\varepsilon_i^2) = 0$.
- $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ as the drawn samples are i.i.d.

Gauss Markov Theorem

Consider the model

$$\underline{Y} = X\beta + \varepsilon,$$

where

$$\underline{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & X_{11} & X_{12} & \cdots & X_{1k} \\ 1 & X_{21} & X_{22} & \cdots & X_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{nk} \end{bmatrix},$$
$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

Then the ordinary least squares estimator $\hat{\beta}$ is the best linear unbiased estimator of β .

Proof of the Gauss-Markov Theorem

We will show $\hat{\beta}$ is an unbiased estimator of β where

$$\hat{\beta} = (X^T X)^{-1} X^T \underline{Y}$$

Let $Z = (X^T X)^{-1} X^T$, then $\hat{\beta} = Z \underline{Y}$.

Now,

$$\begin{aligned} E(\hat{\beta}) &= E(Z \underline{Y}) \\ &= E_{X, \underline{Y}}(Z \underline{Y}) \\ &= E_X(E_{\underline{Y}|X}(Z \underline{Y})) \\ &= E_X\left(Z E_{\underline{Y}|X} \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}\right) \end{aligned}$$

where

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik} + \varepsilon_i$$

and $E(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$

Proof of the Gauss-Markov Theorem (continued)

$$\begin{aligned} E(\hat{\beta}) &= E_X \left(Z \begin{bmatrix} E_{Y|X}(Y_1) \\ E_{Y|X}(Y_2) \\ \vdots \\ E_{Y|X}(Y_n) \end{bmatrix} \right) \\ &= E_X \left(Z \begin{bmatrix} \beta_0 + \beta_1 X_{11} + \cdots + \beta_k X_{1k} \\ \beta_0 + \beta_1 X_{21} + \cdots + \beta_k X_{2k} \\ \vdots \\ \beta_0 + \beta_1 X_{n1} + \cdots + \beta_k X_{nk} \end{bmatrix} \right) \\ &= E_X (ZX\beta) \\ &= \beta E_X (ZX) \\ &= \beta E_X \{ (X^\top X)^{-1} X^\top X \} \\ &= \beta \end{aligned}$$

Proof of the Gauss-Markov Theorem (continued)

Let $\tilde{\beta} = C\underline{Y}$ be another linear estimator of β with $C = (X^\top X)^{-1}X^\top + D$, where D is a $k \times n$ non-zero matrix. As we're restricting to unbiased estimators, minimum mean squared error implies minimum variance. The goal is therefore to show that such an estimator has a variance no smaller than that of $\hat{\beta}$, the OLS estimator.

Proof of the Gauss-Markov Theorem (continued)

We calculate,

$$\begin{aligned} E(\tilde{\beta}) &= E[C\underline{Y}] \\ &= E[((X^\top X)^{-1}X^\top + D)(X\beta + \varepsilon)] \\ &= ((X^\top X)^{-1}X^\top + D)X\beta + ((X^\top X)^{-1}X^\top + D)E[\varepsilon] \\ &= ((X^\top X)^{-1}X^\top + D)X\beta \\ &= ((X^\top X)^{-1}X^\top)X\beta + DX\beta \\ &= (I_k + DX)\beta \end{aligned}$$

Since β is unobservable, $\tilde{\beta}$, is unbiased iff $DX = 0$.

Proof of the Gauss-Markov Theorem (continued)

Now

$$\begin{aligned}\text{Var}(\tilde{\beta}) &= \text{Var}(C\underline{Y}) \\ &= C\text{Var}(\underline{Y})C^\top \\ &= \sigma^2 CC^\top \\ &= \sigma^2((X^\top X)^{-1}X^\top + D)(X(X^\top X)^{-1} + D^\top) \\ &= \sigma^2((X^\top X)^{-1}X^\top X(X^\top X)^{-1} + (X^\top X)^{-1}X^\top D^\top + DX(X^\top X)^{-1} + DD^\top) \\ &= \sigma^2(X^\top X)^{-1} + \sigma^2(X^\top X)^{-1}(DX)^\top + \sigma^2 DX(X^\top X)^{-1} + \sigma^2 DD^\top \\ &= \sigma^2(X^\top X)^{-1} + \sigma^2 DD^\top \quad [DX = 0] \\ &= \text{Var}(\hat{\beta}) + \sigma^2 DD^\top \quad [\sigma^2(X^\top X)^{-1} = \text{Var}(\hat{\beta})]\end{aligned}$$

Since $DD^\top$ is a positive semidefinite matrix, $\text{Var}(\tilde{\beta})$ exceeds by $\text{Var}(\hat{\beta})$ by a positive semidefinite matrix.

Maximum Likelihood Estimator of Ordinary Least Squares

We assume that ε_i 's ($i = 1, 2, \dots, n$) are independent and identically distributed following a normal distribution

$$\mathcal{N}(0, \sigma^2).$$

Now we use the method of maximum likelihood to estimate the parameters of the linear regression model

$$Y_i = a + bX_i + \varepsilon_i, \quad (i = 1, 2, \dots, n),$$

the observations Y_i ($i = 1, 2, \dots, n$) are independently distributed with

$$\mathcal{N}(a + bX_i, \sigma^2)$$

for all $i = 1, 2, \dots, n$.

Maximum Likelihood Estimator of Simple Linear Regression (continued)

The likelihood function of the given observations (X_i, Y_i) and unknown parameters a, b and σ^2 is

$$L(X_i, Y_i; a, b, \sigma^2) = \prod_{i=1}^n \left(\frac{1}{2\pi\sigma^2} \right)^{1/2} \exp \left[-\frac{1}{2\sigma^2} (Y_i - a - bX_i)^2 \right].$$

The maximum likelihood estimates of a, b and σ^2 can be obtained by maximizing $L(X_i, Y_i; a, b, \sigma^2)$ or equivalently its log-likelihood function $\ln L(X_i, Y_i; a, b, \sigma^2)$, where

$$\ln L(X_i, Y_i; a, b, \sigma^2) = -\left(\frac{n}{2}\right) \ln(2\pi) - \left(\frac{n}{2}\right) \ln(\sigma^2) - \left(\frac{1}{2\sigma^2}\right) \sum_{i=1}^n (Y_i - a - bX_i)^2.$$

Maximum Likelihood Estimator of Simple Linear Regression (continued)

The normal equations are obtained by partial differentiation of the log-likelihood with respect to a , b , and σ^2 , and equating them to zero as follows:

$$\frac{\partial \ln L(X_i, Y_i; a, b, \sigma^2)}{\partial a} = -\frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - a - bX_i) = 0$$

$$\frac{\partial \ln L(X_i, Y_i; a, b, \sigma^2)}{\partial b} = -\frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - a - bX_i)X_i = 0$$

and

$$\frac{\partial \ln L(X_i, Y_i; a, b, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (Y_i - a - bX_i)^2 = 0.$$

Maximum Likelihood Estimator of Simple Linear Regression (continued)

The solution of these normal equations gives the maximum likelihood estimates of a , b , and σ^2 as

$$\hat{a} = \bar{Y} - \hat{b}\bar{X},$$
$$\hat{b} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{s_{XY}}{s_{XX}},$$

and

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (Y_i - \hat{a} - \hat{b}X_i)^2}{n},$$

respectively. It can be verified that the Hessian matrix of second-order partial derivatives ensures that the likelihood function is maximized at these values.

Maximum likelihood estimation for multiple linear regression

Theorem: Given a *linear regression model* with correlated observations

$$\underline{Y} = X\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_n), \quad (1)$$

the *maximum likelihood estimates* of β and σ^2 are given by

$$\hat{\beta} = (X^T X)^{-1} X^T \underline{Y}$$

and

$$\hat{\sigma}^2 = \frac{1}{n} (\underline{Y} - X\hat{\beta})^T (\underline{Y} - X\hat{\beta}). \quad (2)$$

where I_n is the identity matrix of order n .

Maximum likelihood estimation for multiple linear regression (continued)

Proof: With the *probability density function* of the multivariate normal distribution, the linear regression equation (1) implies the following likelihood function

$$\begin{aligned} p(\underline{Y} \mid \beta, \sigma^2) &= \mathcal{N}(\underline{Y}; X\beta, \sigma^2 I_n) \\ &= \frac{1}{(2\pi)^{\frac{n}{2}} \sqrt{|\sigma^2 I_n|}} \exp \left[-\frac{1}{2} (\underline{Y} - X\beta)^\top (\sigma^2 I_n)^{-1} (\underline{Y} - X\beta) \right] \end{aligned} \quad (3)$$

Here

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

Using $|\sigma^2 I_n| = (\sigma^2)^n |I_n|$, the log-likelihood function

$$\begin{aligned} l(\beta, \sigma^2) &= \log p(\underline{Y} \mid \beta, \sigma^2) \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2} \log |I_n| - \frac{1}{2\sigma^2} (\underline{Y} - X\beta)^\top I_n^{-1} (\underline{Y} - X\beta). \end{aligned} \quad (4)$$

Maximum likelihood estimation for multiple linear regression (continued)

Substituting the precision matrix $P = I_n^{-1} = I_n$ into (4) to ease notation, we have:

$$l(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2} \log(|I_n|) - \frac{1}{2\sigma^2} (\underline{Y}^\top P \underline{Y} - 2\beta^\top X^\top P \underline{Y} + \beta^\top X^\top P X \beta). \quad (5)$$

The derivative of the log-likelihood function (5) with respect to β is

$$\begin{aligned} \frac{dl(\beta, \sigma^2)}{d\beta} &= \frac{d}{d\beta} \left(-\frac{1}{2\sigma^2} (\underline{Y}^\top P \underline{Y} - 2\beta^\top X^\top P \underline{Y} + \beta^\top X^\top P X \beta) \right) \\ &= \frac{1}{2\sigma^2} \frac{d}{d\beta} (2\beta^\top X^\top P \underline{Y} - \beta^\top X^\top P X \beta) \\ &= \frac{1}{2\sigma^2} (2X^\top P \underline{Y} - 2X^\top P X \beta) \\ &= \frac{1}{\sigma^2} (X^\top P \underline{Y} - X^\top P X \beta). \end{aligned} \quad (6)$$

Maximum likelihood estimation for multiple linear regression (continued)

and setting this derivative to zero gives the MLE for β :

$$\frac{dl(\beta, \sigma^2)}{d\beta} = 0$$

$$0 = \frac{1}{\sigma^2} (X^\top P \underline{Y} - X^\top P X \beta)$$

$$0 = X^\top P \underline{Y} - X^\top P X \beta$$

$$X^\top P X \beta = X^\top P \underline{Y}$$

i.e.,

$$\hat{\beta} = (X^\top P X)^{-1} X^\top P \underline{Y} \quad (7)$$

Maximum likelihood estimation for multiple linear regression (continued)

The derivative of the log-likelihood function (4) at $\hat{\beta}$ with respect to σ^2 is

$$\begin{aligned}\frac{dl(\beta, \sigma^2)}{d\sigma^2} &= \frac{d}{d\sigma^2} \left(-\frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} (\underline{Y} - X\hat{\beta})^\top (\underline{Y} - X\hat{\beta}) \right) \\ &= -\frac{n}{2} \cdot \frac{1}{\sigma^2} + \frac{1}{2(\sigma^2)^2} (\underline{Y} - X\hat{\beta})^\top (\underline{Y} - X\hat{\beta}) \\ &= -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} (\underline{Y} - X\hat{\beta})^\top (\underline{Y} - X\hat{\beta})\end{aligned}\tag{8}$$

Maximum likelihood estimation for multiple linear regression (continued)

and setting this derivative to zero gives the MLE for σ^2 :

$$\begin{aligned}\frac{dl(\beta, \sigma^2)}{d\sigma^2} &= 0 \\ 0 &= -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}(\underline{Y} - X\hat{\beta})^\top(\underline{Y} - X\hat{\beta}) \\ \frac{n}{2\sigma^2} &= \frac{1}{2(\sigma^2)^2}(\underline{Y} - X\hat{\beta})^\top(\underline{Y} - X\hat{\beta}) \\ 2(\sigma^2)^2 \cdot \frac{n}{2\sigma^2} &= 2(\sigma^2)^2 \cdot \frac{1}{2(\sigma^2)^2}(\underline{Y} - X\hat{\beta})^\top(\underline{Y} - X\hat{\beta})\end{aligned}$$

i.e.,

$$\hat{\sigma}^2 = \frac{1}{n}(\underline{Y} - X\hat{\beta})^\top(\underline{Y} - X\hat{\beta}) \quad (9)$$

Together, (7) and (9) constitute the MLE for multiple linear regression.

Testing of hypotheses for slope parameter

Case 1: When σ^2 is known

Consider the simple linear regression model

$$Y_i = a + bX_i + \varepsilon_i \quad (i = 1, 2, \dots, n).$$

It is assumed that the ε_i 's are independent and identically distributed and follow $\mathcal{N}(0, \sigma^2)$.

First, we develop a test for the null hypothesis related to the slope parameter

$$H_0 : b = 0$$

Testing of hypotheses for slope parameter (continued)

Assuming σ^2 to be known, we know that

$$E(\hat{b}) = b, \quad \text{Var}(\hat{b}) = \frac{\sigma^2}{s_{XX}}$$

where $s_{XX} = \sum_{i=1}^n (X_i - \bar{X})^2$ and $\hat{b}$ is a linear combination of normally distributed Y_i 's. So

$$\hat{b} \sim \mathcal{N}\left(b, \frac{\sigma^2}{s_{XX}}\right)$$

and so the following statistic can be constructed

$$Z_1 = \frac{\hat{b} - b}{\sqrt{\frac{\sigma^2}{s_{XX}}}}$$

which is distributed as $\mathcal{N}(0, 1)$ when H_0 is true.

Testing of hypotheses for slope parameter (continued)

A decision rule to test

$$H_1 : b \neq 0$$

can be framed as follows:

$$\text{Reject } H_0 \text{ if } |Z_1| > Z_{\alpha/2}$$

where $Z_{\alpha/2}$ is the $\alpha/2$ percent point on the normal distribution.

Similarly, the decision rule for one-sided alternative hypothesis can also be framed.

Testing of hypotheses for slope parameter (continued)

Case 2: When σ^2 is unknown:

When σ^2 is unknown then we proceed as follows. We know that

$$\frac{SS_{\text{res}}}{\sigma^2} \sim \chi^2(n-2)$$

and

$$E\left(\frac{SS_{\text{res}}}{n-2}\right) = \sigma^2.$$

where $SS_{\text{res}} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$

Testing of hypotheses slope parameter (continued)

Further, $\frac{SS_{\text{res}}}{\sigma^2}$ and b_1 are independently distributed. This result will be proved formally later in the next module on multiple linear regression.

This result also follows from the fact that under normal distribution, the maximum likelihood estimates, viz., the sample mean (estimator of population mean) and the sample variance (estimator of population variance) are independently distributed, so $\hat{b}$ and s^2 are also independently distributed.

Thus the following statistic can be constructed:

$$t_0 = \frac{\hat{b} - b}{\sqrt{\frac{\hat{\sigma}^2}{s_{XX}}}} = \frac{\hat{b} - b}{\sqrt{\frac{SS_{\text{res}}}{(n-2)s_{XX}}}}$$

which follows a t -distribution with $(n-2)$ degrees of freedom, denoted as t_{n-2} , when H_0 is true, here $s_{XX} = \sum_{i=1}^n (X_i - \bar{X})^2$.

Testing of hypotheses for intercept term

Case 1: When σ^2 is known

Suppose the null hypothesis under consideration is

$$H_0 : a = a_0,$$

where σ^2 is known. Then, using the result that

$$E(\hat{a}) = a, \quad \text{Var}(\hat{a}) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{s_{XX}} \right)$$

and a is a linear combination of normally distributed random variables, the following statistic

$$Z_0 = \frac{\hat{a} - a_0}{\sqrt{\sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{s_{XX}} \right)}}$$

has a $\mathcal{N}(0, 1)$ distribution when H_0 is true.

Testing of hypotheses for intercept term (continued)

A decision rule to test

$$H_1 : a \neq a_0$$

can be framed as follows:

$$\text{Reject } H_0 \text{ if } |Z_0| > Z_{\alpha/2}$$

where $Z_{\alpha/2}$ is the $\alpha/2$ percentage point on the normal distribution. Similarly, the decision rule for one-sided alternative hypothesis can also be framed.

Testing of hypotheses for intercept term (continued)

Case 2: When σ^2 is unknown

When σ^2 is unknown, then the following statistic is constructed

$$t_0 = \frac{\hat{a} - a_0}{\sqrt{\frac{SS_{\text{res}}}{n-2} \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{XX}} \right)}}$$

which follows a t -distribution with $(n-2)$ degrees of freedom, i.e., t_{n-2} , when H_0 is true.

A decision rule to test

$$H_1 : a \neq a_0$$

is as follows:

Reject H_0 whenever $|t_0| > t_{n-2, \alpha/2}$

where $t_{n-2, \alpha/2}$ is the $\alpha/2$ percentage point of the t -distribution with $(n-2)$ degrees of freedom. Similarly, the decision rule for one-sided alternative hypothesis can also be framed.

General Linear Hypothesis

Assume

$$Y \sim N(X\beta, \sigma^2 I_n),$$

where Y is $n \times 1$, X is $n \times p$, and β is $p \times 1$. Let $\text{rank}(X) = r$.

Then

$$RSS = \min_{\beta} (Y - X\beta)^{\top} (Y - X\beta),$$

and the minimum is attained at

$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} Y.$$

$$\frac{RSS}{\sigma^2} \sim \chi_{n-r}^2.$$

Law of Iterated Expectation (Vector Form)

For any integrable function $g : \mathbb{R}^{2n} \rightarrow \mathbb{R}$,

$$\mathbb{E}_{\mathbf{X}, \mathbf{Y}}[g(\mathbf{X}, \mathbf{Y})] = \mathbb{E}_{\mathbf{X}}[\mathbb{E}_{\mathbf{Y}|\mathbf{X}}[g(\mathbf{X}, \mathbf{Y})]].$$

Variances

We are assuming that Y_i 's are independently distributed,

$$\begin{aligned}\text{Var}(\hat{b}) &= \sum_{i=1}^n k_i^2 \text{Var}(Y_i) + \sum_i \sum_{j \neq i} k_i k_j \text{Cov}(Y_i, Y_j) \\ &= \sigma^2 \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{s_{XX}^2} \\ &= \frac{\sigma^2 s_{XX}}{s_{XX}^2} \\ &= \frac{\sigma^2}{s_{XX}}\end{aligned}$$

Here we have used the fact that $\text{Cov}(Y_i, Y_j) = 0$ as $Y_1, \dots, Y_n$ are independent.

Variances (continued)

Also

$$\text{Var}(\hat{a}) = \text{Var}(\bar{Y}) + \bar{X}^2 \text{Var}(\hat{b}) - 2\bar{X} \text{Cov}(\bar{Y}, \hat{b})$$

Now,

$$\begin{aligned}\text{Cov}(\bar{Y}, \hat{b}) &= E[\{\bar{Y} - E(\bar{Y})\}\{\hat{b} - E(\hat{b})\}] \\&= E\left[\bar{\varepsilon}\left(\sum_{i=1}^n k_i Y_i - b\right)\right] \\&= \frac{1}{n} E\left[\left(\sum_{i=1}^n \varepsilon_i\right)\left(a \sum_{i=1}^n k_i + b \sum_{i=1}^n c_i X_i + \sum_{i=1}^n k_i \varepsilon_i\right) - b \sum_{i=1}^n \varepsilon_i\right] \\&= \frac{1}{n} [0 + 0 + 0 + 0] \\&= 0\end{aligned}$$

So,

$$\mathbf{Var}(\hat{a}) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{s_{XX}} \right)$$

References

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Thank you

